

THE FAÇADE OF
NEWTONIANISM
In Networking and its Consequences

Paul Borrill, Work in Progress 02023-Dec-04a, Updated Nov-06

Abstract

It's not what we don't know that gets us into trouble, it's what we know for certain that just ain't so. Computer scientists 'make believe' that time is Newtonian; this assumption, in their software, networks and protocols, manifests in a belief that clocks *can* be *synchronized*, and *time flows the same everywhere*. This article is in response to the recent Open Compute conference in San Jose, and a reminder of what physicists already know about the nature of time, and where proven physics is may suggest that the "Time Appliance Project" (TAP) may be barking up the wrong tree. Any attempt to create a single "imaginary" timeline against which we can determine the order of events is destined to fail, and the result will be lost and corrupted data.

CAUSALITY, MONOTONICITY & TIME

TIME IS RELATIVE, SO IS EVENT ORDER

Different observers in a network will see events in a different time-order. We don't require spacetime intervals on a Minkowski Manifold to prove this. The existence of a 'finite' speed of light (SOL) is sufficient.

Newtonian Time assumes an *infinite* speed of light. Measuring the one-way speed of light is fraught with issues of clock synchronization. The SOL can only be measured by a round trip ([it's impossible to measure the one-way SOL](#)).

Everything we measure is from the past and discrete. The "photon causality assumption" implies that the photon was sent by the emitter 'before' it was received by the absorber.

Photons don't have a before and after. Proper time from the photon's perspective is zero, no matter how far it travels.

This unfortunate assumption is reflected in Lamport's "happened before" (hb) principle, which implies an infinitely accurate guillotine and is missing the counterfactual "happened exactly" (he) which is how a round-trip protocol can "synchronize" because it creates "logical Simultaneity".



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TIME CANNOT BE SYNCHRONIZED

But it can be disseminated

A common misconception conflates the distribution (*dissemination*) of time, and the *synchronization* of time. These are independent concepts, and present different theoretical difficulties. The dissemination of time is in principle implementable in GPSS Navigation and Time. The *synchronization* of Time is not, and represents a belief in simultaneity (Newtonian time). Einstein showed us that simultaneity is a myth.

We cannot synchronize time in principle [SR], it will therefore be problematic in practice.

In a follow-on article, we will show how it's possible to 'engineer' a necessary and sufficient alternative to Minkowski simultaneity in an FPGA substructure, as long as the number of bits, spatial representation in a register, and the speed of the clock (≈ 1 GHz) are compatible. These '1ns synchronization structures' form the roots of a DAG in a causal network. Once we have this tool in our toolbox, distributed systems engineers can get to work on going beyond repetition and redundancy, to provide more 9's, which is never enough, to a more useful 'determinism on demand'.

We misunderstand (and misinterpret) how to do transactions in conventional Processors with a Cache Hierarchy. Unless we 'pre-manage' the [cache lines](#), we won't get low latency transactions.

We talk about time synchronization, in the application of GPSS, but we are really talking about frequency synchronization, which requires a relatively long integration period, to get time.

TIME IS NON-MONOTONIC

Monotonicity implies a scalar - a single total order for everyone in the universe, and denies both special and general relativity.

If the physics model of spacetime cannot be monotonic, then neither will be the event structure in an asynchronous distributed system. There is no [homomorphism](#) (structure preserving map) connecting the two models.

Logical Monotonicity is inconsistent with physics, and will fail when [indefinite causality](#) reorders events. We will find it impossible to understand this in the context of irreversible time (i.e. insisting on Monotonicity). We can, however, understand it within the context of Indefinite Causal Order. Like Bell states in Quantum Theory, these results occur over any arbitrary distance and are not confined to atomic scales. We see indefinite causality within a rack, or across a datacenter.

CAUSALITY IS INDEFINITE

We know from Quantum mechanics that uncertainty in the product of time and energy revolve around the Planck's constant. We know from physics that it takes an infinite amount of energy to be infinitely accurate in time. $\Delta E \Delta T \leq \frac{h}{2\pi}$. See later section: we cannot avoid the physics, and Quantum Clock.

This is in addition to the other major result in physics that time and space revolve around the speed of light, c .

Three of the most important discoveries since Einstein, are the verification of Bell state measurements in Entanglement (Nobel Prize 2022), Attosecond pulses (Nobel Prize 2023), and the decisive validation of Indefinite Causality (a future Nobel Prize?).

Entrenched beliefs in Newtonian time cannot survive the light of experimental scrutiny. Newtonian Time, and the synchronizability of clocks is a falsifiable theory, and it has been falsified.

HARNESSING INDEFINITE CAUSALITY

[Indefinite causality](#) cannot be overcome by repetition (bang bang bang networking with timeouts and retries). It cannot be overcome by stuffing packets "in streaming mode" into a network. More bandwidth on separate paths won't help either, if each of them is stuffing bits as fast as they can on each path. It cannot be overcome by redundancy (multipathing and gossip protocols just make things worse, and precipitate cascade failures which will result in catastrophic data loss and corruption).

These conventional mitigations in the datacenter will lead to Limpware, Gray Failure and Metastable Failure. Even at small scale; corruptions & data loss happen regularly in small clusters.

TIME DOES NOT FLOW LIKE NEWTON

Most computer scientists assume that time flows constantly everywhere, and is irreversible. In physics and real world systems, the truth is far stranger.

Computer Scientists can no longer ignore what physicists know about the nature of time. The machine code of the universe is quantum mechanics. It behaves in superpositions, not as black or white zeros or ones. What observers measure is not the time a packet is sent on the network, it's the time of receipt. And a lot can happen between sending and receiving, especially in a network that drops, delays, duplicates and reorders messages.

SIMULTANEITY IS A MYTH

In (1949) Einstein wrote:

"There is no such thing as simultaneity of distant events; consequently, there is also no such thing as immediate action at a distance in the sense of Newtonian mechanics. Although the introduction of actions at a distance, which propagate at the speed of light, remains feasible according to this theory, it appears unnatural; for in such a theory there could be no reasonable expression for the principle of conservation of energy. It therefore appears unavoidable that physical reality must be described in terms of continuous functions in space."

[Up to Diffeomorphism](#), Reflections on Relativity.

TIMESTAMPS ARE AN UNSAFE ORDERING CONSTRUCT

All timestamps, whether they are (frequency) synchronized to NTP, PTP, White Rabbit, or some future Attosecond mechanism, are a delusion of Newtonian Time thinking.

Attempting to create a global order will always fail, and those failures will show up in lost and corrupted data. We see this all the time in, Dropbox, iCloud, OneDrive and gDrive.

This doesn't mean that we can't create "local" islands of "Partial Order". We know how to do that in a single machine. Even if there are atomicity failures in a single core, preempted by the Linux Kernel, we know how to "reverse" those locally (under the hood) to heal their "illusion" of monotonicity.

Timestamps (physical, logical) deny both Special Relativity and General Relativity. Once you have a finite speed of light, you have a conundrum: Either everyone else is in *your* past (in your positive time), or you are in everyone

else's past — in your negative time (which looks like influence from the future) if we continue to insist on Newtonian time. Some physicists call this retro causality. Others see it as simply the symmetry of time in Quantum Information Theory.

Spacetime is not Newtonian. Computer scientists trying to force it to be Newtonian will run into the brick wall of lost and corrupted data. Assuming that time is smooth, irreversible (and monotonic). Is what we call the Minkowski Error. Modern physics doesn't work this way. The Time Appliance Project (TAP) is barking up the wrong tree.

TIME IS NOT LIKE A RACE

Almost all depictions of time in computer science look like a race, with the runners (observers) moving down their 'lanes' and arriving at some pre-defined finish line. The leader or coordinator starts the race and determines when it ends. Sometimes when the last runner crosses the finish line; sometimes when a quorum $2f + 1$ completes, with timeouts to restart the stragglers if one or more of them get lost. We even call it a 'race condition'.

IS TIME MORE LIKE A PING-PONG GAME?

But what if time is *not a race*? What if it's more like a game of ping-pong? With each element of the game played between exactly two players (plus an umpire who keeps score).

A match is played best 3 of 5 games (or 4/7 or 5/9). For each game, the first player to reach 11 points wins that game, however a game must be won by at least a two point margin. A rally is like a round of consensus.

The Ping Pong Ball is a conserved quantity (CQ). What comes over the net goes back over the net. Failing to return the ball means losing points. Throughout the game it is possible for the umpire (and observers) to make sure they are all accounted for.

By managing operations as conserved quantities (tokens or ping pong balls) we can now engineer high resolution accounting. This is not possible if the protocol throws away the bipartite symmetry of the interaction. Entanglement is monogamous for a reason.

Conserved quantities is compositional across many links. Under failure conditions, the ball is on one side of the net or the other. There are no timeouts. There is no playing against the clock like in chess. The symmetry of the events (server and responder) takes care of 'liveness', and the umpire takes care of 'safety'.

PING PONG TIME IS SYMMETRIC

We break the initial symmetry by declaring one side to be the server, Alice, or Bob are now entanglement partners.

IT TAKES TWO TO TANGO, AND THREE TO PARTY

BELIEF IN CLOCK SYNTONIZATION

Is no longer tenable in modern datacenter design. It will lead to lost and corrupted data structures, and Subsecond Tsunamis.

SUBSECOND TSUNAMIS AND DELAYS IN DECENTRALIZED ELECTRONIC SYSTEMS

Pedro D. Manrique. [MDPI Download](#). Extreme (subsecond tsunami) behaviors are regularly observed in electronic exchanges. A few actors can game the system by triggering an extreme movement, both upwards and downwards in the movement of stock prices. One of the deep failures of the electronic exchanges is their reliance on clock synchronization. Equinix offers equal length cables to their customers in an attempt to make things "look" fair, but the unfairness comes from a misunderstanding about the nature of time, and engineering and artificial relationship to a Newtonian simultaneity plane only makes it easier to game the system.

USING SYNCHRONIZED CLOCKS TO AVOID COORDINATION WILL EXACERBATE FAILURES

There are many attempts in the distributed systems literature to use both tightly as well as loosely synchronized clocks to avoid round-trip costs in consensus protocols. ePaxos, Nezha, Accord, and many other protocols have followed this path.

This attempt to create an "imaginary" timeline against which we can determine the order of events is destined to fail and the result will be lost and corrupted data. Different observers will necessarily register different partial orders; there is and can never be a total order for all observers, unless those total orders are synchronized by protocols in which one of them is the "causal leader". This is implicitly what we do in State Machine Replication (SMR) and leader-based consensus protocols.

Clocks cannot be synchronized, they can only be disseminated in a "tree" of Bell (entanglement-like correlations). Signaling requires a two-way exchange at the speed of light.

ATOMICITY FAILURES ARE EVERYWHERE

In Processors

In Applications

In the Kernel

In the SmartNIC

In Fsync

In the Filesystem

In Storage

.... But Not In the DAE FPGA link...

SIMULTANEITY PLANES ARE A DELUSION

Simultaneity is Meaningless without *Interaction*

Minkowski Space is flat when it contains no matter or energy.

ZIMULTANEITY

Is a bipartite (monogamous) thing, just like entanglement.

The concept of simultaneity is meaningless without Interaction, and every interaction requires 'pairs' of back to back uninterrupted messages (No blocking or stuttering allowed).

Zimultaneity is what we can synthesize within an FPGA Substructure as one SmartNIC takes to another over an Ethernet Link. It doesn't rely on the fools errand of trying to synchronize a set of free-running clocks to some non-existent 'big clock in the sky'.

MONOTONICITY: MISSING HALF THE STORY

Monotonicity is missing half the story. Counterfactuals in our views of time. We only see time going forwards because when time goes backwards, we forget.

Memory is essential to our notion of time. There is no past if we don't remember it.

LOGICAL TIME: ILLUSION OR DELUSION?

Happens before or Unhappens before?

You can't have happened before without happened exactly.

Time is change that we can count.

-If we can't count its, not real.

Monotonicity isn't counting, it's smash and restart between islands snapshots.

We can only count something locally under our own control.

SIMULTANEITY PLANES ARE A DELUSION

PHYSICAL TIME

Sharpening the Guillotine

It doesn't matter if our time intervals are seconds (PPS), milliseconds (NTP), Microseconds (PTP), Nanoseconds (WW) Femtoseconds or Attoseconds. We still run into the same problem: timestamps are an unsafe ordering construct.

When the sign of time (+/-) is necessarily opposite between entities in an entangled system(s), then our Newtonian assumption that it moves positively forward uniformly (and everywhere) the same, is in conflict with logic, physics and the experimental record.

A POSSIBLE SOLUTION

To create Simultaneity without clocks. Synchronize a set of 64 bits on the same clock edge as a slice comes into an FPGA. By using the mathematics of entanglement we can "mix" the bits that only Alice can set, and the bits that only Bob can set, and subject them to [rewriting rules](#), which creates a cellular automata "simple rules" local communication. No clocks required.

At DÆDÆLUS we have invented a radical way to exploit this new scientific understanding, and designed it in to the Ethernet links we use for communication between servers in datacenters. And thereby uniquely solving fundamental distributed systems problems, such as: robust failure detectors, secure consensus, reversible vector clocks which enable *hypertransactions*.

MULLIGAN STEW - MARK BURGESS

In Mark Burgess & Andreás Gerlits' non-blocking data store,

COMMENTS DURING MEETING

Mark: "Time basically unfolds when we observe change not guess that change might have happened when we see something change, that's a tick of a clock."

Time is measured by the observer, not when the same signal was sent. It's when the signal was received. And that that's what confuses everybody, they think the more accuracy will be better. Relying on the

Edward: I want to try to clarify the way that I use the term timestamp. Okay. I don't use the word timestamp to represent some mythical accurate measure of some mythical physical time. That's not the way I use it. The way that I use it is that if you're going to have consensus about the state of a system, you have to have agreement on the order in which changes to that state

happened. That's where timestamps come in. timestamps are about agreeing on the order in which changes to state happen.

clock synchronization error is indistinguishable from network latency. Of course, you actually can't tell the difference between the two, it's physically impossible to tell the difference. Consequently, if you have a system where speed of light constraints tell you that your network latency is going to be greater than 100 nanoseconds, there's really no point in getting picosecond clock accuracy. It doesn't buy you anything, right? Because the latency and network latency is going to totally dominate the clock accuracy problem.

Yeah, I think I think I get what what Mark is saying, but I wouldn't express it that way because I don't think the external world appears to be going slower. If you have a, you know, a quasar that is, you know, light years away that is oscillating at a 100 megahertz, it appears to be oscillating at 100 megahertz. And it appears that time is progressing extremely rapidly there. So. So I think the way that I would express what you're, I think saying, which I agree with, is that when you want to have a consistent view of what is happening across a system, that's when things have to slow down. And it's only when you have that bidirectional causality. Okay? If you if you only have unidirectional causality, it's not a problem. So the database world I think, keeps confusing things because databases fundamentally always have this notion of having to have a global consistency. But databases are a subset of systems of interest, I would say some set of measures zero. So they're either useful, but they're not that by any that snarky comment to be distributed systems, there are many useful distributed systems that don't have that strong consistency requirements.

Andreas: This was exactly the design decision that led to the to the interleaving base clock, that there's no point in, in progressing time, in any quicker weigh down the minimum network distance between to do so there's no point in being quicker than the network distance between two observers. Otto two participants in this case,

AFTER THE MEETING ON EMAIL FROM MARK

I wanted to clarify my possibly clumsy explanation of why interior/exterior time see a mismatch of time rates (time slow etc). I think the essence was as Edward described it, but it missed some points. I don't know if this is better, but let's try.

- Each agent observes at the maximum rate of its own proper time process. It can't see anything faster by Nyquist's theorem. There is a triangle relation for what's required for two agents to experience equivalent time rates.
- Interior time is that time rate of the agent experiences by virtue of its own interior processes. e.g the program counter in a program
- Exterior time is measured by the changes that happens between the agent and other agents,

e.g. transactions completed

- Because several interior operations are required for a single exterior cycle of change to complete between agents, an exterior clock will tick less often than the interior clock. On the exterior clock always proceeds at the same constant rate (e.g. speed of light). One cycle per time unit.
- If an agent observes that on its interior clock, more ticks/ time units will have passed on the interior per single exterior cycle/tick, so transactional time seems to slower. This is exactly what Einstein relativity is about, but without the continuum geometry.
- This is not about distance. When you compare what happens on Mars with what happens on Earth, both events are exterior events to the observer, so they may happen as fast or as slow, even though the transactions are delayed. They are both compared on the same clock, so there's no contradiction, But the round trip time is a slow process compared to the changes observed.
- Agents with large interior component systems will typically have up to N^2 interactions to complete a cycle for the whole agent to complete a change (e.g. consensus or coordination) so many interior ticks are required for the whole agent to respond to an exterior interaction. Bigger agent may be up to N^2 slower depending on how much interior interaction (e.g. sharing or consensus) is needed.

The watched pot never seems to boil because thoughts race when waiting, so an agent experiences many accelerated thoughts/ticks while the kettle boils at a constant exterior time rate. If the waiting agent is busy, it will tend to measure relative to longer interior cycles and won't notice the difference. These rates are captured in brain waves, alpha, beta, gamma etc.

btw, the squishy messy latency Pat referred to is basically equivalent to curved spacetime in Einsteinian physics. So it does happen there too.

RESPONSE FROM EDWARD

I would also like to clarify why I think the Shannon-Nyquist theorem is a red herring.

This theorem is really about ambiguity, not about limits in rates.

If a signal is sampled at 10 samples per second, for example, and what is observed appears to be a 2 Hz signal, it may actually be a 12 Hz signal, or a 22 Hz signal, or 2,000,002 Hz signal. The observed signal is the same in all these cases. It is quite common to undersample signals when the remote rate of change is approximately known.

For example, a digital radio can pick up a 5 GHz radio signal without sampling it at 10 GHz (which would be extremely expensive). What matters is the bandwidth of the signal, not

the underlying rate of change. If the signal is limited to the range 4,999,999,995 Hz and 5,000,000,005 Hz, then sampling it at 10 Hz will be fast enough to unambiguously extract it.

MARK's REPLY

Hi Edward,

Yes, this is true.

The Nyquist rate is relevant but not in the way you thought I meant.

I did not argue that the sampling rate affects the rate of system change, only that it affects the certainly with which change can be discerned reliably and therefore deterministically. This is particularly true for discrete events. E.g. If a security camera sweeps twice a minute, a burglar can run across the yard without being seen if he's quick enough, while it's pointing in the opposite direction. This is the connection for observation of a purely discrete event process.

The effective rate of *proper* time, on the other hand, is a scaling effect of dependent processes and thus of clocks (scaling in the sense of the renormalization group, not related to distance or latency). It's like subroutines in a program. Subroutines complete quicker than the program as a whole, and individual program steps are executed more quickly than subroutines complete and return functional values. The interior rate of subroutine execution is faster than the exterior rate of function evaluation. Express that for any spacetime bounded process.

So we are both not wrong! Hope this makes sense

MONOTONIC POSITIONING (FROM PAT)

It's not completely the same, though, because time is monotonic even across the two, and it's aligning those monotonic observations, I would think so when I think of consensus, I think of going to a random place to a random place and trying to align, compare and swap kind of things. So this is just a question of what the word consensus means. That's all.

Yeah, I'm just saying that time has a monotonicity, which is a narrower, less complex problem, than substituting random order from outside things that are competing to change the next value.

THE DANGER OF ATOMIC OPERATIONS

Even the most well thought out theories in computer science are rife with assumptions that are just not true in practice.

Advice from [Abseil](#) (open source C++ libraries drawn from the most fundamental pieces of Google's internal codebase):

"Most engineers reach for atomic operations in an attempt to produce some lock-free mechanism. Furthermore, programmers enjoy the intellectual puzzle of using atomic operations. Both of these lead to clever implementations which are almost always ill-advised and often incorrect.

Algorithms involving atomic operations are extremely subtle. For example, discovering a general-purpose, efficient, lock-free, singly-linked list algorithm took significant research and required care to implement.

Almost all programmers make mistakes when they attempt direct use of atomic operations. Even when they don't make mistakes, the resulting code is hard for others to maintain.

Atomic operations should be used only in a handful of low-level data structures ... written by a few experts and then reviewed and tested thoroughly.

Unfortunately, many people attempt to write lock-free code, and this is almost always a mistake.

Please do not fall into this trap: do not use atomic operations. If you do, you will make mistakes, and those will cost the owners of that code time and money.

There are a number of existing higher-level components that are already carefully crafted, reviewed, and tested. Use them if they do what you need. Otherwise, use mutexes."

We can't trust anything in the Host processor (e.g. C++ standard library on Von Neumann cores) or conventional storage systems, to be *atomic*. At DÆDÆLUS, all our atomicity mechanisms are implemented in FGPAs (in SmartNICs now, ASICs in future).

DETERMINISM ON DEMAND

In a world where certain resources are abundant (e.g. Bisection Bandwidth) and other resources are scarce (e.g. Cost, Power, Latency, GPU Utilization), can we do a tradeoff by using a model of time to overcome the [Attack of the Killer Microseconds](#)? [Taming the Killer Microsecond](#). Need to say something about cost and power..

Can we use flexible quorums to better manage the schedulability of software systems so that they don't pay such a large penalty for the nondeterminism inherent in current networks?.

Time is not Newtonian, Smooth, or Irreversible.

Any attempt to create an "imaginary" timeline against which we can determine the order of events is destined to fail and the result will be lost and corrupted data. This is exactly what we've seen happening in all distributed databases for several decades.

A more detailed problem statement might be:

- Maintaining liveness and synchronizing processes in networks is a challenge when packets can be dropped, reordered, duplicated or delayed

- Conventional networks require protocol stacks and applications to use timeouts and retries to maintain liveness
- This makes exactly-once semantics impossible and precipitates retry storms which lead to unbounded tail latency and transaction failure
- We've known since 1980 that [mutual inconsistency](#) results in silent corruption and loss of data.

The computer science community have made no progress because they continue to think of time as Newtonian, and attempt to measure intervals (and causality) as if it were on a smooth, irreversible manifold.

WE CAN'T AVOID THE PHYSICS

The canonical article used to disabuse developers of their misunderstandings about the nature of time in distributed systems is Justin Sheehy's "[There is no now](#)".

This is on the edge of readability for people who are stuck in their belief systems. Justin doesn't mince words when he tells people that they are wrong or just plain silly.

Unfortunately this is still somewhat simplistic from a physics perspective. It only tells half the story from a relativity perspective and says absolutely nothing about the Quantum mechanical universe that we live in. At least he is treated as an authority from a developer perspective and debunks the usual nonsense about Google Spanner.

A deeper perspective on the relationship of time in physics and computer science can be found in the Videos by Edward Lee:

1. [Verifying Parallel and Distributed Systems: The Observer Problem.](#)
2. [Reactors and Lingua Franca: A Programming Model for Cyberphysical Systems.](#) Synchronized clocks and Timestamps.

Relationship between Energy and Time.

Wikipedia on Heisenberg's Uncertainty principle. This is central to many common misunderstandings about time:

"Other than the position–momentum uncertainty relation, the most important uncertainty relation is that between energy and time. The energy–time uncertainty relation is not, however, an obvious consequence of the general Robertson–Schrödinger relation. Since energy bears the same relation to time as momentum does to space in special relativity, it was clear to many early founders, Niels Bohr among them, that the following relation should hold: $\Delta E \Delta t \gtrsim \hbar$

but it was not always obvious what Δt precisely meant. The problem is that the time at which the particle has a given state is not an operator belonging to the particle, it is a parameter describing the evolution of the system.

As Lev Landau once joked "To violate the time–energy uncertainty relation all I have to do is measure the energy very precisely and then look at my watch!"

One false formulation of the energy-time uncertainty principle says that measuring the energy of a quantum system to an accuracy ΔE requires a time interval $\Delta t > \frac{h}{\Delta E}$.

This formulation is similar to the one alluded to in Landau's joke, and was explicitly invalidated by Y. Aharonov and D. Bohm in 1961.

The time Δt in the uncertainty relation is the time during which the system exists unperturbed, not the time during which the experimental equipment is turned on, whereas the position in the other version of the principle refers to where the particle has some probability to be and not where the observer might look.

Another common misconception is that the energy–time uncertainty principle says that the conservation of energy can be temporarily violated—energy can be "borrowed" from the universe as long as it is "returned" within a short amount of time. Although this agrees with the spirit of relativistic quantum mechanics, it is based on the false axiom that the energy of the universe is an exactly known parameter at all times. More accurately, when events transpire at shorter time intervals, there is a greater uncertainty in the energy of these events.

Therefore it is not that the conservation of energy is violated when quantum field theory uses temporary electron–positron pairs in its calculations"

but that *the energy of quantum systems is not known with enough precision to limit their behavior to a single, simple history.*

Thus *the influence of all histories* must be incorporated into quantum calculations, including those with much greater or much less energy than the mean of the measured/calculated energy distribution.

I believe this ties in directly with Mark's intuition of how to "weave the rope" in our discussion yesterday.

GOING DEEPER — QUANTUM CLOCKS

This notion of inner clocks and outer clocks is related to the long-standing issue on the status of time in quantum mechanics.

[From [X/Twitter](#) by [Stefano Gogioso](#)]

"The status of time in quantum mechanics has been controversial since the early days of the field [1-4].

In 1926-1928, Dirac, Heisenberg and Bohr considered time as a quantum observable, conjugate to energy and subject with it to various forms of the uncertainty principle. This sounded rather sensible, until Schrödinger argued in 1931 that if the quantum time observable was measured by observing an ideal quantum clock then the energy of a quantum system would become completely uncertain, a condition which he deemed to be "physically meaningless". In 1933, Pauli made this claim rigorous, in a result that would henceforth be known as "Pauli's Theorem". Dirac and Heisenberg revised their positions, and time has been treated as *classical parameter* by the majority of quantum mechanics literature ever since.

If we look at the arguments by Schrödinger and Pauli more carefully, we quickly spot a problem: on the one hand, they are measuring the *clock time* observable, but on the other hand they are talking about the *system energy* observable. If we look at a synchronized clock system state (cf. Figures (1) and (2) below), it's pretty easy to see that the energy measurement on the system and the time measurement for the clock commute, since they are performed at different places. The apparent physical absurdity arises from an incorrect identification of two distinct energy observables: (i) the *clock energy* observable, relevant for the quantum systems governed by the clock and conjugate to the clock time observable; (ii) the physical Hamiltonian for the clock itself, which relates to some other, larger clock and which will not be affected by the clock time measurement.

It might now be tempting to simply posit the existence of some large, universal quantum clock ticking time for all quantum clocks in the universe, but it seems that either: (A) the clock is accessible, in which case we can measure and which freezes time for the entire universe (what nonsense!); (B) the clock is external to the theory, in which case time is a parameter and we are back to square 1. Luckily, there's option (C): we can talk about quantum dynamics based on hierarchies of (locally) synchronized quantum clocks, which can be made inaccessible as desired by fixing total energy constraints. This gives a toy model of emergent global time, without any need for an accessible universal quantum clock [5].

The key step of this approach is to show how quantum clocks arise from ordinary quantum systems, a process which is sketched in Figure (3) below. We consider two (or more) quantum systems governed by a large "outer" clock, which we make inaccessible by imposing a total energy constraint (the outer clock time becomes fully uncertain this way). Under certain conditions, one (or more) of the synchronized systems (marked by V in the Figure below) possesses an observable conjugate to its Hamiltonian (related to the "outer" clock): this is our long sought after time observable, and it can be used to turn the system into a smaller "inner" clock which governs all remaining quantum systems.

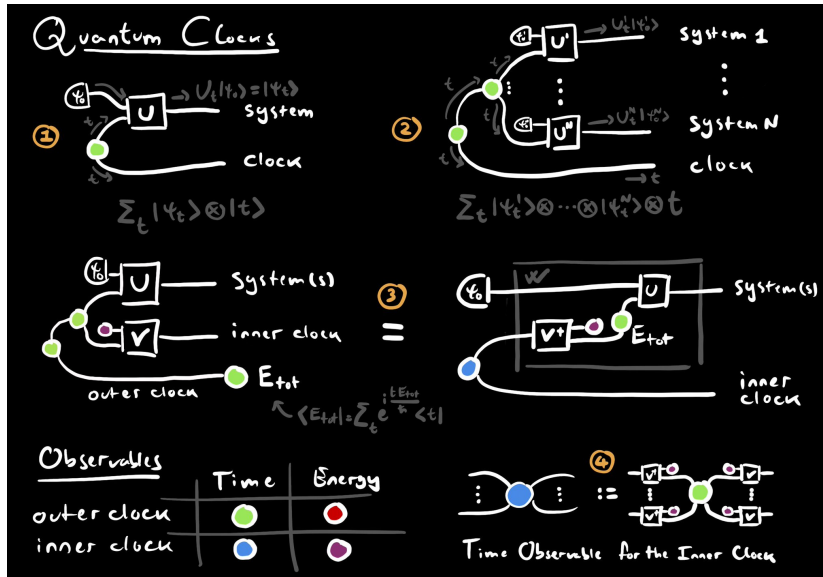
Now imagine that our original outer clock arises from some larger outer clock, perhaps together with other sibling clocks. And then that the larger clock itself arises from even larger clocks above it, and so on, in a hierarchy where time grows finer as quantum clocks grow larger. In the limit, this paints a picture of what a consistent treatment of universal quantum time might look like... but that's a picture for another day."

TIME APPLIANCE PROJECT IS BARKING UP THE WRONG TREE.

Is a Causality Appliance Project (CAP) more of what we need? (Based on Lingua Franca)? Can the CAL registers be accommodated in the SmartNICs (This can be tested with FPGAs).

RTT BASED CONGESTION CONTROL IS BARKING UP THE WRONG TREE

"New technology sometimes presents an opportunity to revisit how we design systems by changing our basic underlying assumptions. This is the story of this paper, which challenges some conventional wisdom about delay not being an accurate congestion signal in low latency networks by using recent advances in NIC hardware."



This is Newtonian Time thinking.

GOOGLE IS BARKING UP THE WRONG TREE

The preferred programming methodology at Google is Synchronous programming [Attack of the Killer Microseconds](#). Although there are various ways we can engineer synchronization in our systems, this always requires round trips. Attempts to do it with one way signals (dissemination) are destined to fail. When our entire compute hierarchy is focussed around being synchronous with DRAM Memory (10-100ns), the pace of the machine will be governed 10-100 Mhz, at least an order of magnitude lower than the time it takes to access a register. Temporal intimacy for a host processor is on the order of a time it takes to load a whole cache line, if you are lucky, and on the order of time it takes to load a new page, if you're not. And if that doesn't bite you then the Linux scheduler will get you in the 1-10 millisecond range.

CLOCK SYNCHRONIZATION INACCURACY

[From Edward Lee — Mulligan Stew 2023-Nov-03] "Clock synchronization error is indistinguishable from network latency. You actually can't tell the difference between the two. It's physically impossible to tell the difference. Consequently if you have a system where the Speed of Light constraints tell you that Network Latency is going to be greater than 100ns, there's really no point in getting picosecond clock accuracy."

WHAT NIST HAS TO SAY ABOUT TIME

From [Fundamentals of Time and Frequency](#).

Table 17.3 Relationship of Frequency Uncertainty to Time Uncertainty

Frequency Uncertainty	Measurement Period	Time Uncertainty
$\pm 1.00 \times 10^{-3}$	1 s	± 1 ms
$\pm 1.00 \times 10^{-6}$	1 s	± 1 μ s
$\pm 1.00 \times 10^{-9}$	1 s	± 1 ns
$\pm 2.78 \times 10^{-7}$	1 h	± 1 ms
$\pm 2.78 \times 10^{-10}$	1 h	± 1 μ s
$\pm 2.78 \times 10^{-13}$	1 h	± 1 ns
$\pm 1.16 \times 10^{-8}$	1 day	± 1 ms
$\pm 1.16 \times 10^{-11}$	1 day	± 1 μ s
$\pm 1.16 \times 10^{-14}$	1 day	± 1 ns

Stability of the device improving as the averaging period (τ) gets longer, since some noise types can be removed by averaging. At some point, however, more averaging no longer improves the results.

SUMMARY & CONCLUSIONS

Atomicity doesn't scale, so it has to be modular and compositional.

[Get description from discussion with Stefano]

INDETERMINATE SIMULTANEITY PLANES

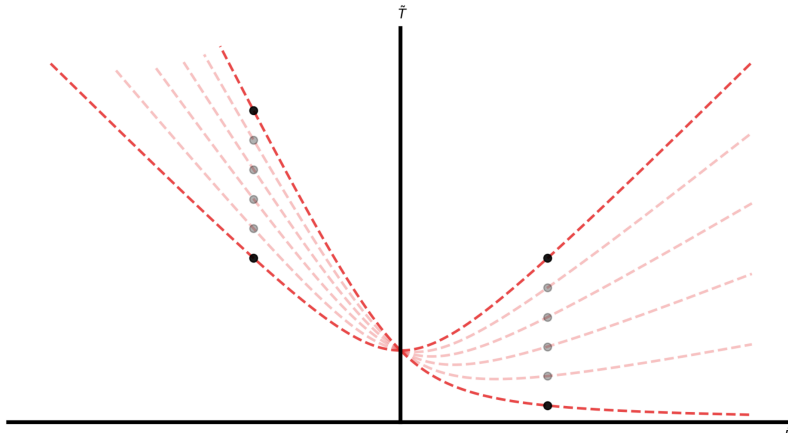


Figure 1: Screenshot 6:19:28 PM.

From: The One-Way Speed of Light and the Milne Universe

The line of simultaneity in the emitter in the FRW coordinates, mapped into the anisotropic velocity of light coordinates from $\kappa = 0$, the isotropic case represented as a hyperbola, to the extreme case, with $\kappa = 1$, both presented in bolder red, with intermediate cases, in steps of $\kappa = 0.2$, presented in lighter red. The filled circles represent the location of the emitter in these coordinates for each of the cases. Note that the spatial location of the emitter in the R coordinate is independent of κ .

JONATHAN OPPENHEIM:

The post-quantum view of Gravity.

[Quanta article.](#)

[Is it Time to Rethink Quantum Gravity?](#)

[Might there be no quantum gravity at all?](#)

END NOTES AND REFERENCES

EndNotes and References

It is necessary to distinguish between the dissemination of time (what we get from GPS satellites) and the synchronization of time (what we do in the IEEE 1588 clock tree hierarchy).

Dissemination of System Time is an old idea. C. Ellingson; R. Kulpinski.1973.

Consider the problem of estimating the offset in timing of like events at geographically separated locations as a basis for establishing common knowledge of time and, hence, system synchronism. Configurations discussed involve interrogation and reply between a user and a **single donor, and one-way propagation between a user and the multiple sites**

of a reference system. The latter category includes navigation systems, which are shown to be appropriate means for disseminating time.

Further ramifications of time dissemination are discussed, including the characteristics of clocks suitable for airborne application. Variables and their relationship and solution are defined for stationary and moving users, and for users of atomic and crystal clocks. For the case in which frequency is offset between clocks, as is likely when crystal oscillators are used, methods are described that permit the estimate of offset in frequency as well as in time. Recursive minimum-variance methods are examined in some detail, and a parametric analysis of performance relative to random and systematic sources of error.

Pauli Erker

1. [Impact of Imperfect Timekeeping on Quantum Control.](#)

“In order to unitarily evolve a quantum system, an agent requires knowledge of time, a parameter that no physical clock can ever perfectly characterize. In this Letter, we study how limitations on acquiring knowledge of time impact controlled quantum operations in different paradigms. We show that the quality of timekeeping an agent has access to limits the circuit complexity they are able to achieve within circuit-based quantum computation. We do this by deriving an upper bound on the average gate fidelity achievable under imperfect timekeeping for a general class of random circuits. Another area where quantum control is relevant is quantum thermodynamics. In that context, we show that cooling a qubit can be achieved using a timer of arbitrary quality for control: timekeeping error only impacts the rate of cooling and not the achievable temperature. Our analysis combines techniques from the study of autonomous quantum clocks and the theory of quantum channels to understand the effect of imperfect timekeeping on controlled quantum dynamics.”

2. [Autonomous Quantum Clocks: Does Thermodynamics Limit Our Ability to Measure Time?](#)

Time remains one of the least well-understood concepts in physics, most notably in quantum mechanics. A central goal is to find the fundamental limits of measuring time. One of the main obstacles is the fact that time is not an observable and thus has to be measured indirectly. Here, we explore these questions by introducing a model of time measurements that is complete and autonomous. Specifically, our autonomous quantum clock consists of a system out of thermal equilibrium—a prerequisite for any system to function as a clock—powered by minimal resources, namely, two thermal baths at different temperatures. Through a detailed analysis of this specific clock model, we find that the laws of thermodynamics dictate a trade-off between the amount of dissipated heat and the clock’s performance in terms of its accuracy and resolution. Our results furthermore imply that a fundamental entropy production is associated with the operation of any autonomous quantum clock, assuming that quantum machines cannot achieve perfect efficiency at finite power. More generally, autonomous clocks provide a natural framework for the exploration of fundamental questions about time in quantum theory and beyond.

3. [The Quantum Hourglass.](#)

In this thesis a general non-relativistic operational model for quantum clocks called quantum hourglass is devised. Taking results from the field of Quantum Information Theory the limitations of such time measurement devices are explored in different physical and information- theoretic scenarios. It will be shown for certain cases that the time resolution of the quantum hourglass is limited by its power consumption.

4. [Quanta Article.](#)

5. [ItsAboutTime.club Podcast with Pauli Erker.](#)

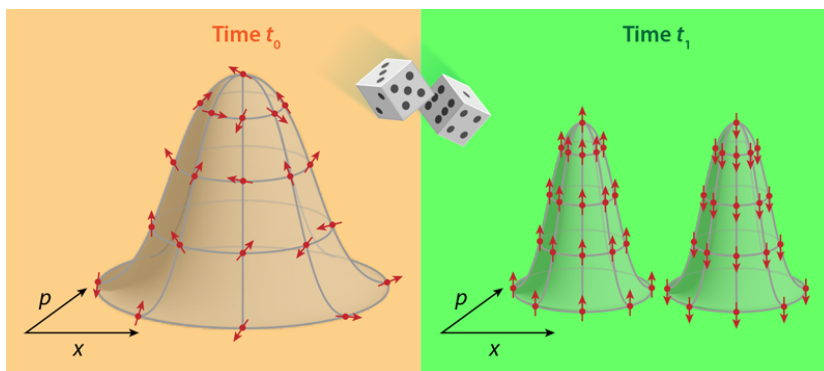
An autonomous quantum clock, is a self-contained device working without any external control or timing. The resources powering the clock should not themselves require another clock to be prepared.

Comment: Entropy is photons. When we go from a hot bath to a colder one, the number of photons from the hot side will (transmitter) will overwhelm the number of photons going from the receiver to the transmitter.

6. [Time symmetry in quantum theories and beyond](#). John H. Selby, Maria Stasinou, Stefano Gogioso, and Bob Coecke:

There is a stark tension among different formulations of quantum theory in that some are fundamentally time-symmetric and others are radically time-asymmetric. This tension is crisply captured when thinking of physical theories as theories of processes. We review process theories and their diagrammatic representation, and show how quantum theory can be described in this language. The tension between time-symmetry and time-asymmetry is then captured by the tension between two of the key structures that are used in this framework. On the one hand, the symmetry is captured by a dagger structure, which is represented by a reflection of diagrams. On the other hand, the asymmetry is captured by a condition involving discarding which, ultimately, is responsible for the theory being compatible with relativistic causality.

Next we consider three different ways in which we this tension can be resolved. The first of these is closely related to recent work of Lucien Hardy, where the tension is resolved by adding in a time reversed version of discarding together with a suitable consistency condition. The second is, to our knowledge, a new approach. Here the tension is resolved by adding in new systems which propagate backwards in time, and imposing a consistency condition to avoid running into well known time-travel paradoxes. The final approach that we explore is closely related to work of Oreshkov and Cerf, where the tension is resolved by removing the constraint associated with discarding. We show two equivalent ways in which this can be done whilst ensuring that the resulting theory still makes sensible operational predictions.



Ľaslav Brukner

Giulia Rubino

[Causality and the arrow of time in the quantum world](#)

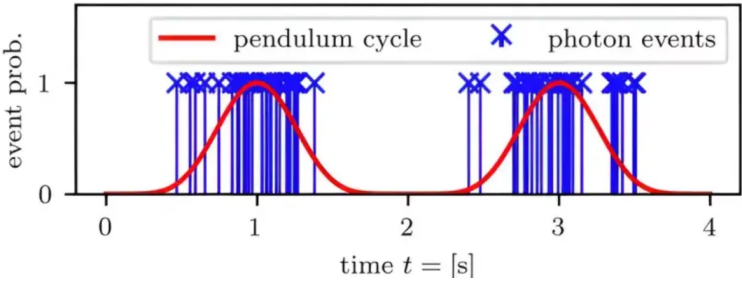
Lucien Hardy

Asynchronous systems at Sun

Ivan Sutherland's [The Tyranny of the Clock](#)

[Limits for Quantum Computers: Perfect Clocks are Impossible, Research Finds](#)

Perfect Clocks are Impossible:



Note to self: Time to put this into Latex